Advanced Materials

Soft Matter 2023

Dr. Ulla Gerling-Driessen

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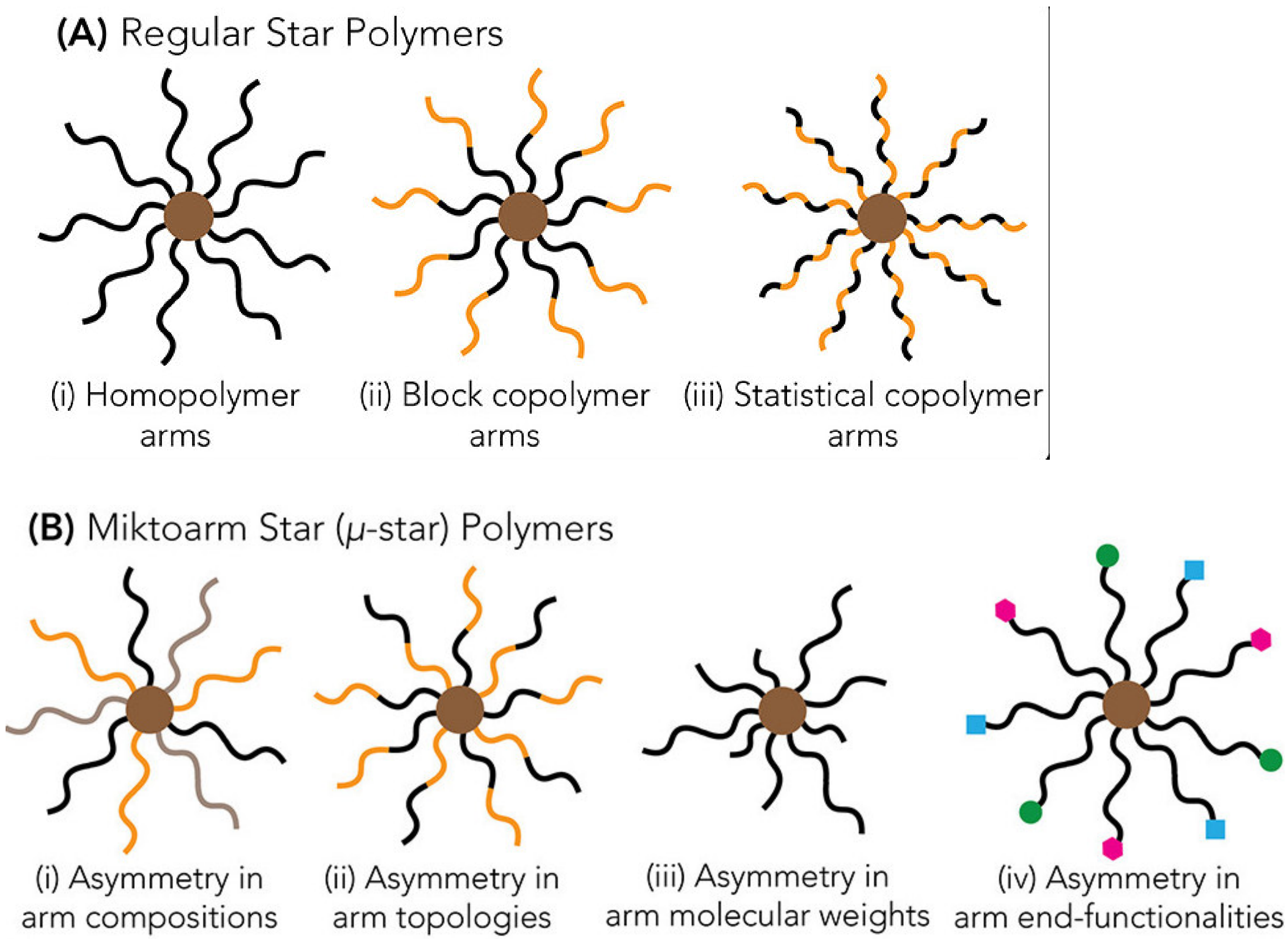
Was sind Miktoarm (Co)Polymere?

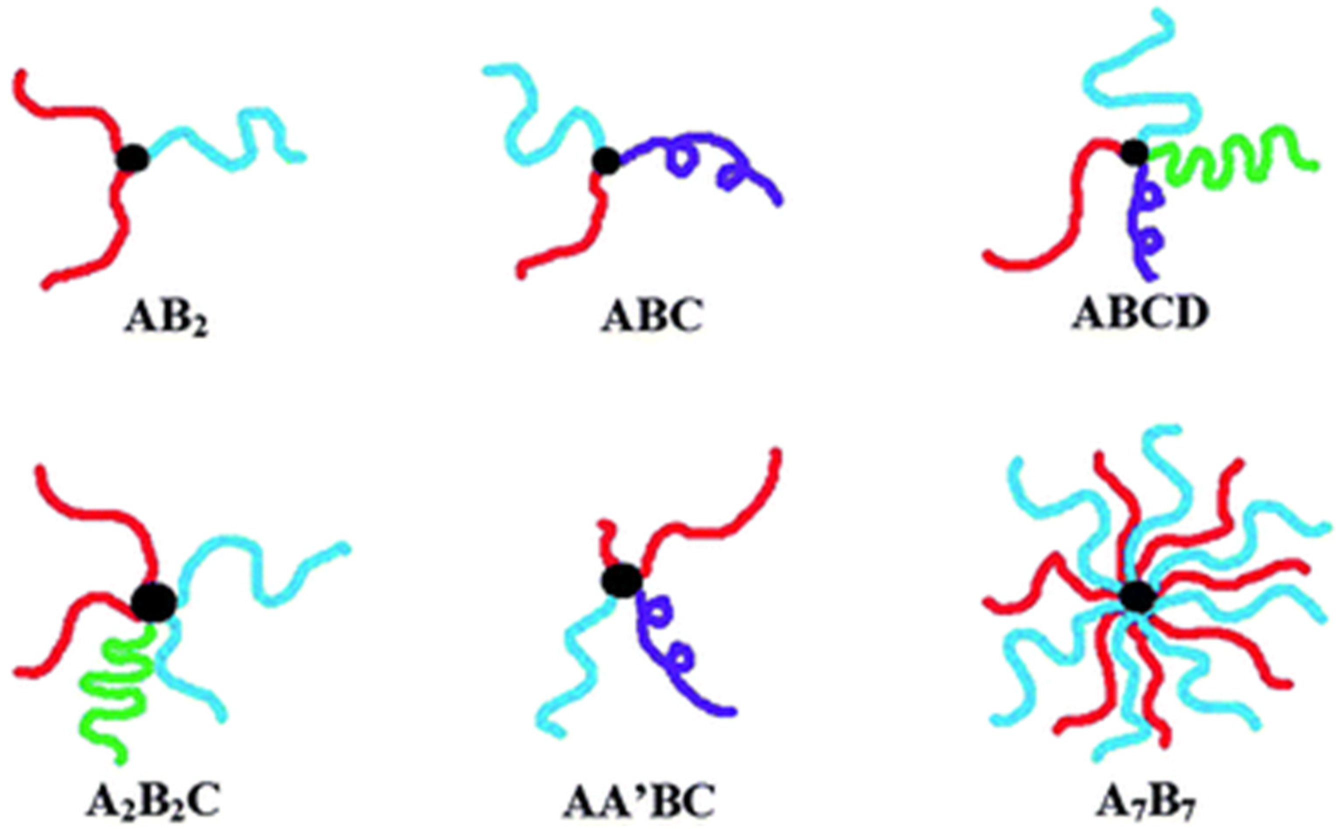


- Miktoarm Polymere (auch μ-star polymere genannt) sind eine besondere Form von Sternpolymeren

- Neue Eigenschaften, die sich z.T. nicht mit linearen Strukturen erhalten werden können

Chem. Mater. 2022, 34, 14, 6188-6209

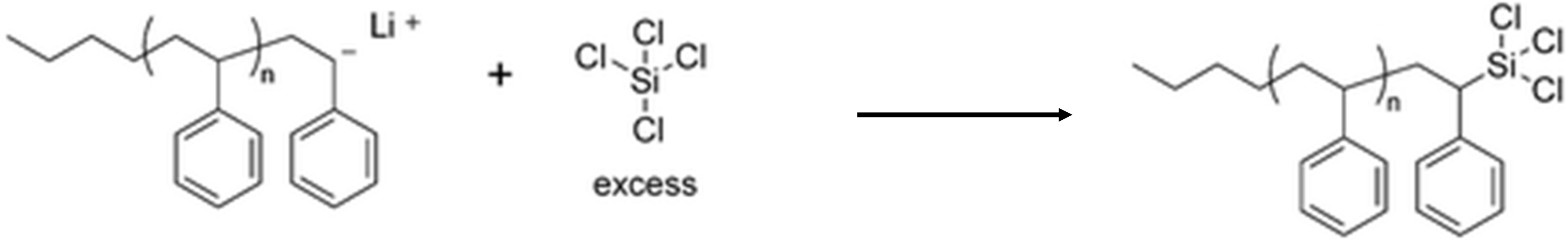
Nomenklatur von Miktoarm (Co)Polymeren



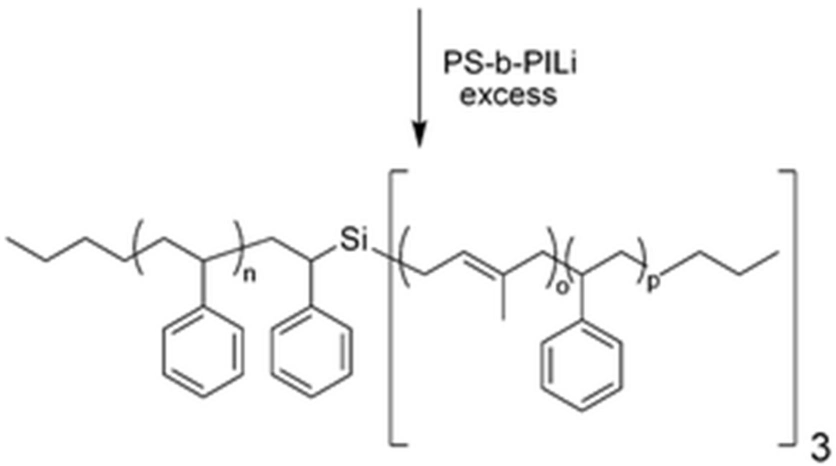
Polym. Chem., 2010, **1**, 1171-1185

Synthese Strategien von Miktoarm (Co)Polymeren

<u>Substitution an Chlor Silane Verbindungen:</u>



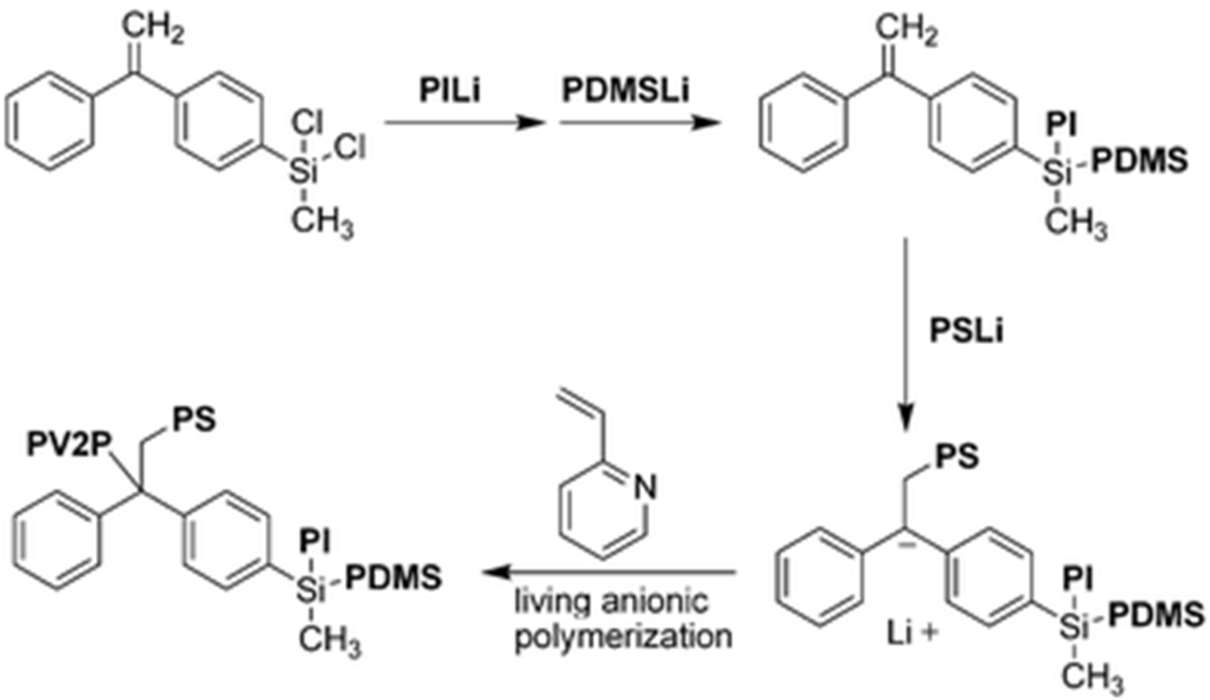
- System: polystyrene (PS) and polystyrene-*b*-polyisoprene
- **A(AB)$_3$** Miktoarm Polymer



- Chlorosilane als core
- Linken von Polymeren mit reaktiven Endgruppen via *living anionic polymerization*
- chlorosilane Verbindung im Übershuss um Mono-Substitution zu gewährleisten

Eur. Polym. J., 2003, **39** , 2155.

Chlor Silane Compunds in Verbindung mit anderen Strathegien

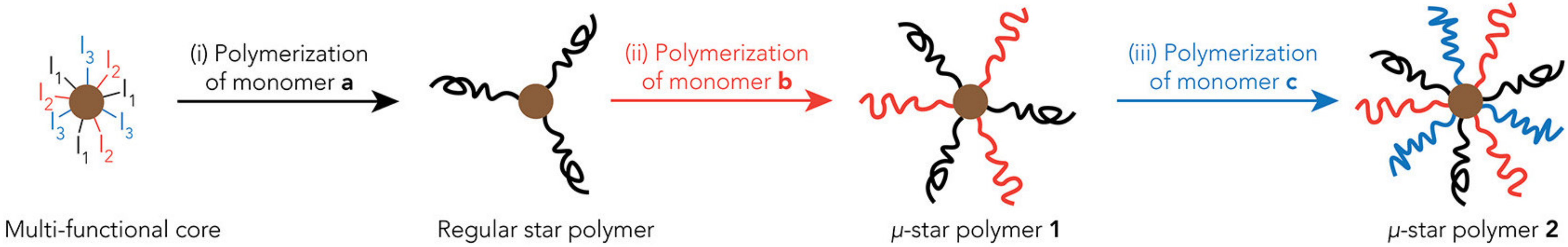


- a single core for the attachment of four different polymers
- **ABCD**-type miktoarm star with four mutually incompatible arms: polystyrene, polyisoprene, polydimethylsiloxane (PDMS), and poly(2-vinylpyridine) (P2VP)

Macromolecules, 2006, **39** , 535.

- core is composed of two chlorosilanegroups and diphenylethylene
- To create the diblock copolymer precursor: sequential titration on the core:
- 1. for monosubstitution of one chlorine with PI
- 2. for the monosubstitution of the second chlorine with PDMS
- living polystyrene chain end reacted on the vinyl group of the diphenylethylene moiety on the core (triblock copolymer).
- Finally, the resulting anion was exploited for the living anionic polymerization of 2-vinylpyridine, (final ABCD miktoarm star polymer)

Core-first Approach



- synthesis of a multifunctional initiator ("multifunctional core,") containing orthogonal initiating sites
- each arm is grown outwards through a combination of different sequential polymerization techniques
- number of arms defined by quantity and type of different initiating sites in the core
- relatively small number of arms (e.g., <10)
- compact core

- common complications:
- protection/deprotection steps
- use of orthogonal polymerization chemistries for different arm (careful selection of monomers)
- side reactions during polymerization (bimolecular termination)

Chem. Mater. 2022, 34, 14, 6188-6209

Core-first Approach: Beispiel

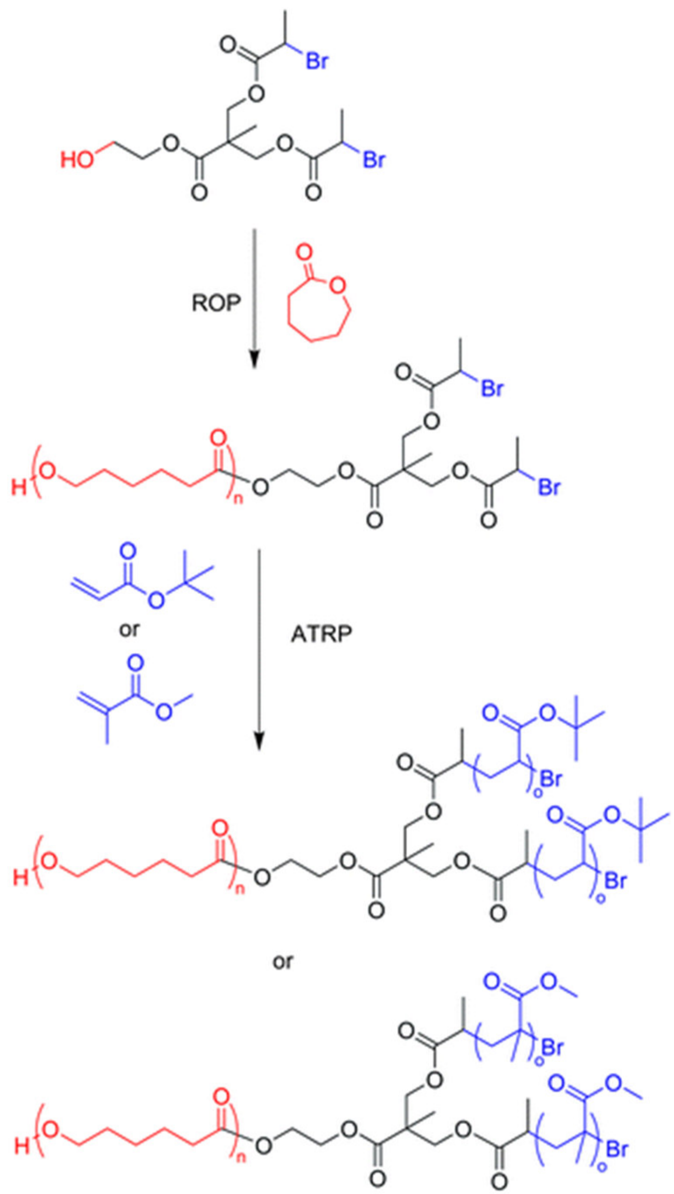
- AB$_2$ miktoarm star polymer:
- first step was the synthesis of a multifunctional initiator designed for sequential ROP and ATRP

- core was used for ROP of ε-caprolactone through the alcohol to generate a polymer functionalized with two bromine end groups
- These sites were polymerized *via* ATRP using either *tert*-butyl acrylate
- or methyl methacrylate

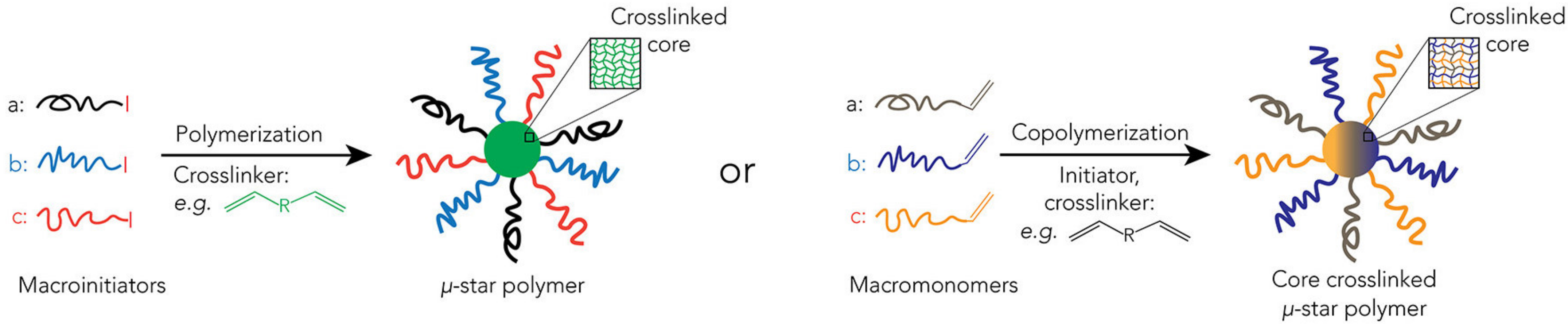
Ring-opening polymerization (ROP)

Atom Transfer Radical Polymerization (ATRP)

J. Polym. Sci., Part A: Polym. Chem., 2004, **42** , 2313.



Arm-first Approach



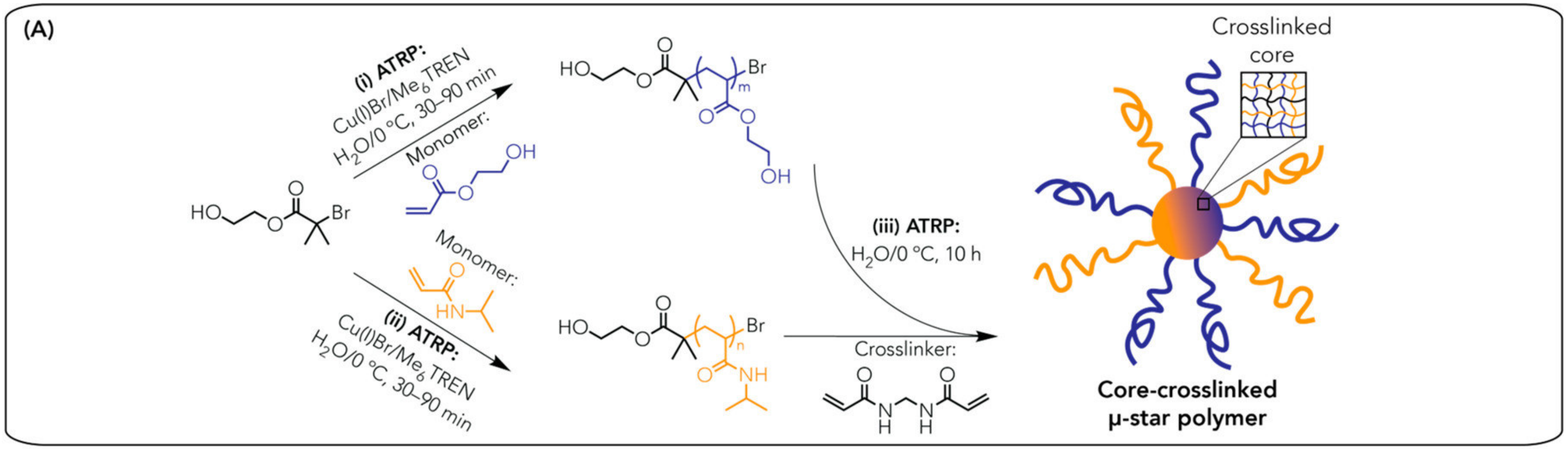
- Generally, in this methodology, the chain ends of many linear macroinitiators, which are formed from a number of CRP methods, are used to polymerize a divinyl compound, typically divinylbenzene (DVB)

- The final miktoarm polymer consisted of a crosslinked microgel core composed of DVB, which ties together the many polymer arms that initiated the DVB polymerization, with the polymers emanating outwards.

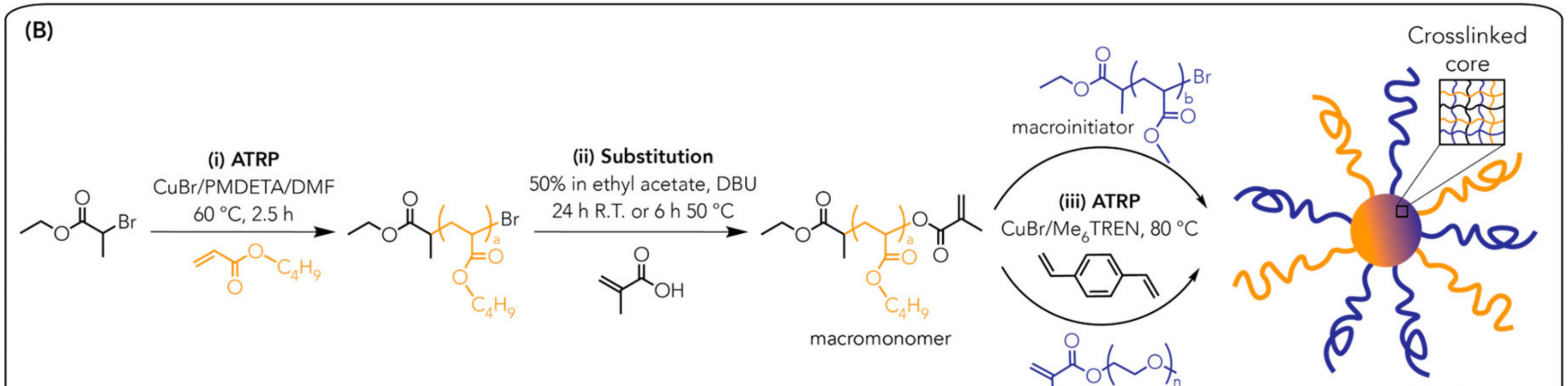
Chem. Mater. 2022, 34, 14, 6188-6209

Arm-first Approach: Beispiel

Macromolecules **2014**, *47* (22), 7869– 7877



- one-pot cross-linking reaction via ATRP
- poly(2-hydroxyethyl acrylate)
- poly(*N*-isopropylacrylamide)



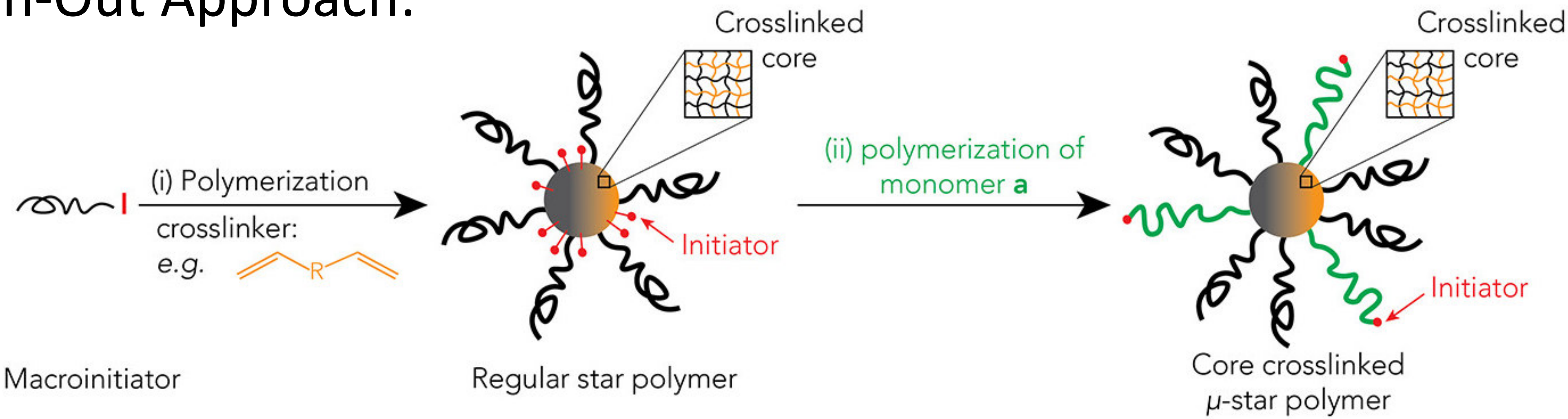
Macromolecules **2008**, *41* (12), 4250– 4257

- cross-linking of
- macroinitiator or
- macromonomer
- or two different macromonomers via ATRP

- common issue in ATRP: loss of chain end functionality, or "livingness"

- partial or full replacement of macroinitiators with macromonomers
- reduces the amount of initiator used for cross-linking
- minimizes termination reactions at the beginning of ATRP

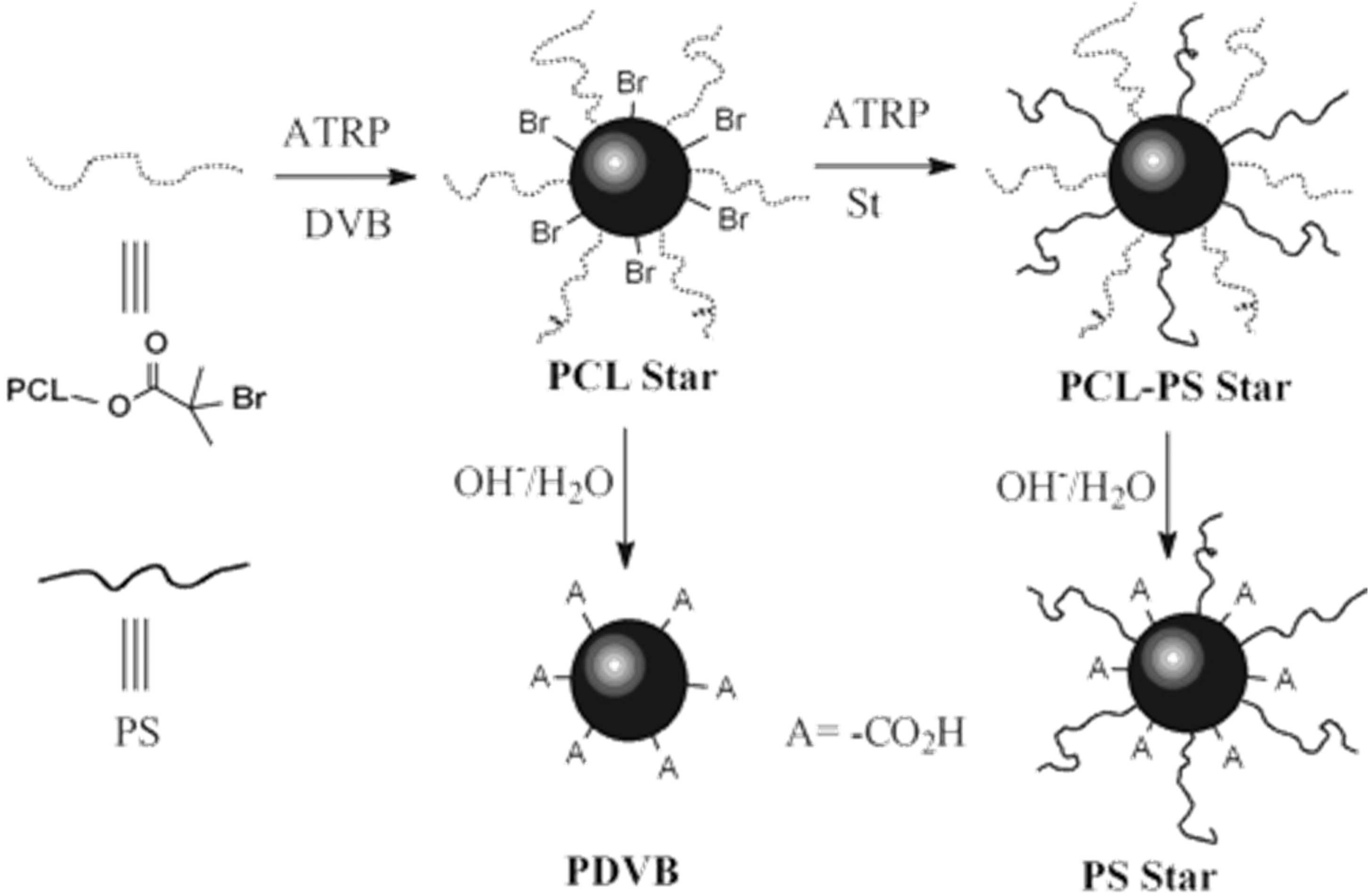
In-Out Approach:



- cross between the “core-first” and “arm-first” methodologies
- a “living” macroinitiator (typically made by CRP methods such as ATRP) initiates the polymerization of a cross-linking agent (such as DVB) to form a homoarm star polymer with polymer arms emanating from the core
- Because the macroinitiator's initiating sites are preserved within the core, these initiating sites are used for subsequent polymerization of a second type of monomer
- Unlike the “core-first” or “arm-first” technique, only A_nB_m-type miktoarm polymers can be built, *i.e.*, only two types of polymer arms can exist in a miktoarm star polymer synthesized by the “in–out” technique
- the number of arms of the polymer grown from the core is always less than the number of arms from the macroinitiator (therefore $m < n$) (***steric hindrance***)

Chem. Mater. 2022, 34, 14, 6188-6209

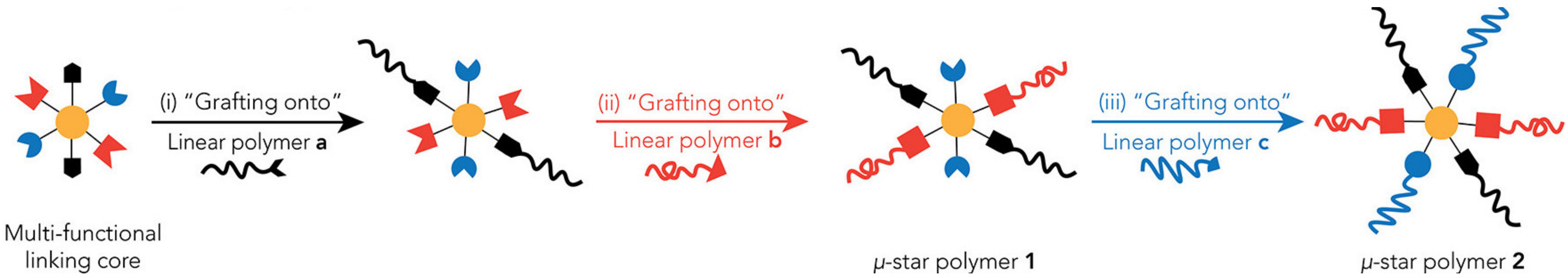
In-Out Approach: Beispiel



- ➢ PCL and PS linked to a DVB microgel core
- ➢ first synthesizing PCL which was end-functionalized with a bromine atom
- ➢ subsequently reacted *via* ATRP with DVB to form a homoarm star polymer macroinitiator
- ➢ bromine atoms remaining on the core were subsequently polymerized *via* ATRP with styrene to form the final miktoarm star polymer.

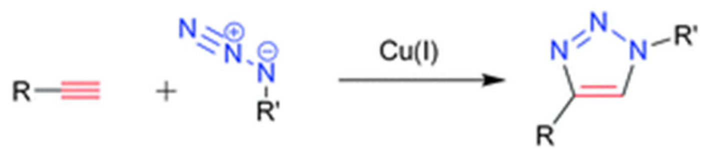
Macromolecules 2004, 37, 10, 3588−3594

Grafting-onto Approach:



Chem. Mater. 2022, 34, 14, 6188-6209

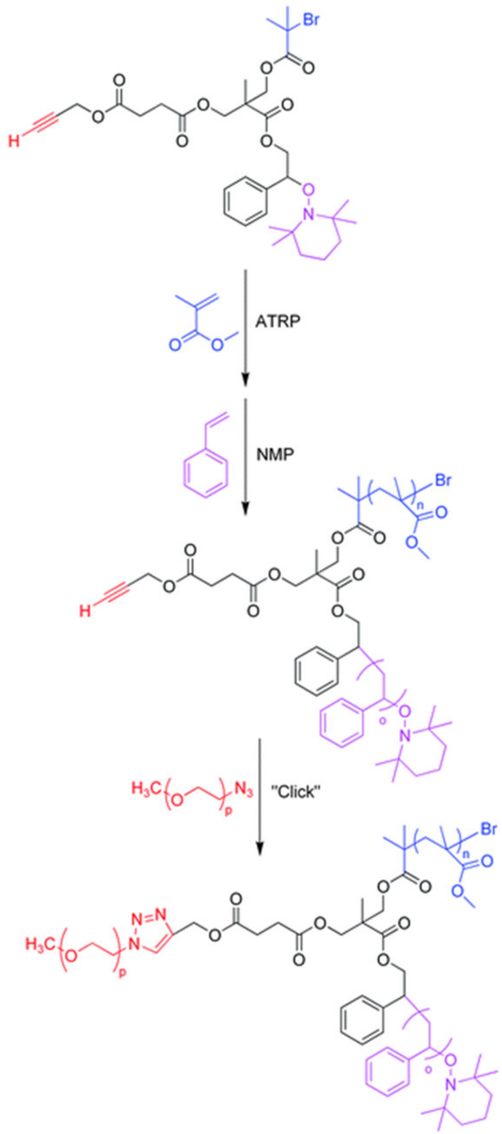
- coupling of the reactive end of at least one polymer arm to a multifunctional core using highly efficient and orthogonal reactions
- remaining arms on a miktoarm polymer are typically grown using controlled polymerization methods (ROP, living anionic polymerization, *etc.)*
- Popularity of the coupling methods can partially be attributed to the rising popularity of "click" chemistry in macromolecular synthesis



Polym. Chem., 2010, **1**, 1171-1185

Grafting-onto Approach: Beispiel

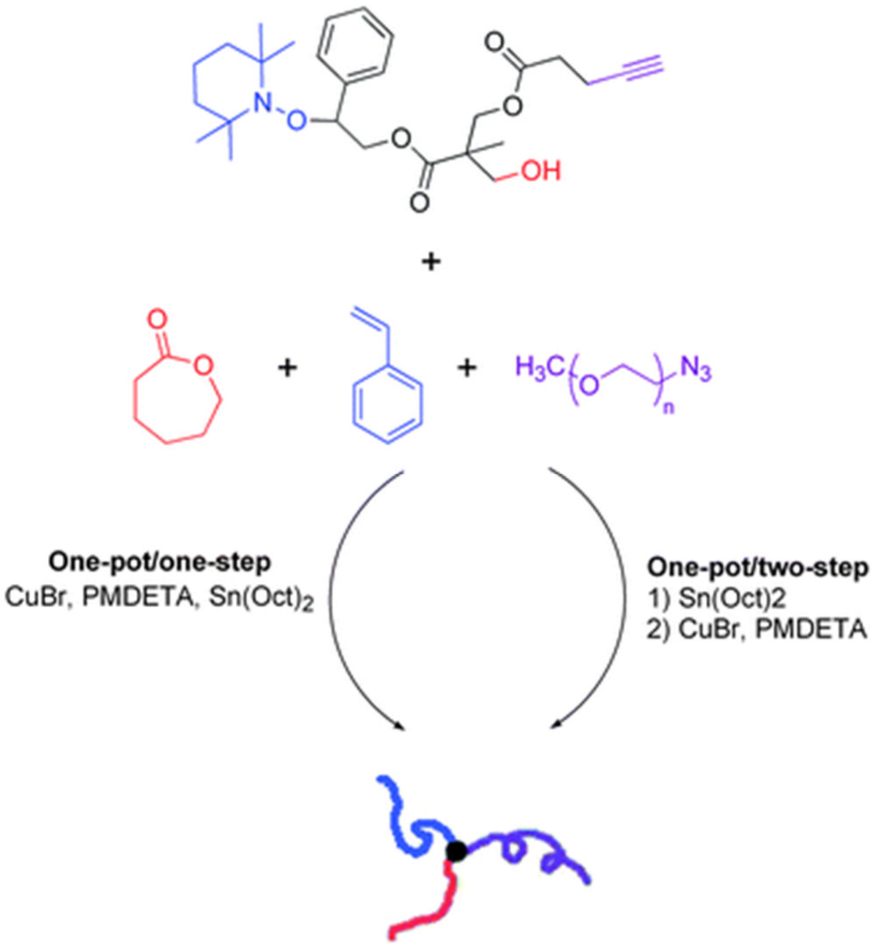
- ATRP and NMP were used to polymerize PMMA and PS respectively from a trifunctional core
- leaving a block copolymer with an alkyne as the only remaining functionality which is located at the junction
- Subsequent “click” reaction with either azide-terminated PEG or PtBA yielded an ABC-type miktoarm star polymer



nitroxide-mediated radical **polymerization** (**NMP**)

J. Polym. Sci., Part A: Polym. Chem., 2006, **44** , 5699

Grafting-onto Approach: Beispiel

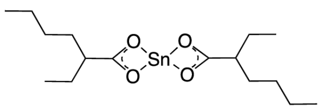


Polym. Sci., Part A: Polym. Chem., 2007, **45** , 3588

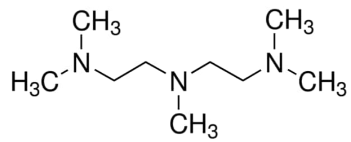
- Combined “click” reaction between an azide and an alkyne with ROP and ATRP
- ABC miktoarm polymer composed of:
  - Polycaprolacton (PCL)
  - poly((2-dimethylamino)ethyl methacrylate) (PDMA)
  - either PS or PEG in a one-pot fashion

- The trifunctional core, ε-caprolactone monomer, 2-(dimethylamino)ethyl acrylate monomer, and an azide-terminated polymer (PS or PEG) were mixed together with catalystst in(II) 2-ethylhexanoate (Sn(Oct)$_2$), copper(I) bromide (CuBr), and *N*,*N*,*N*′,*N*′,*N*″-pentamethyldiethylenetriamine (CuBr/PMDETA) and stirred at 80 °C for 12 hours

- Sn(Oct)$_2$ works as a catalyst for ROP
- CuBr/PMDETA doubles as a catalyst for both ATRP and the “click” reaction

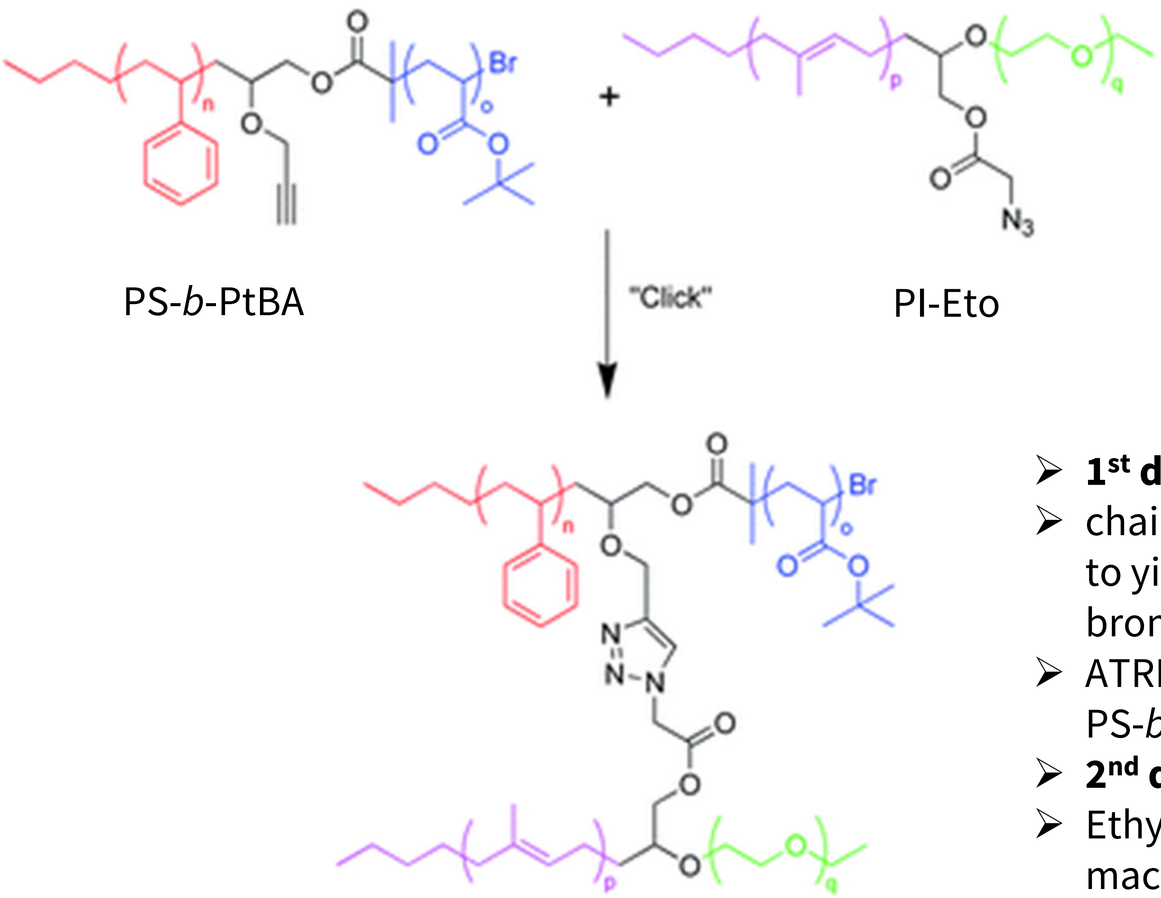


Sn(Oct)2 (Zinn(**II**)-**2**-ethylhexanoat)



N,N,N′,N′′,N′′-Pentamethyldiethylentriamin

Grafting-onto Approach: Beispiel

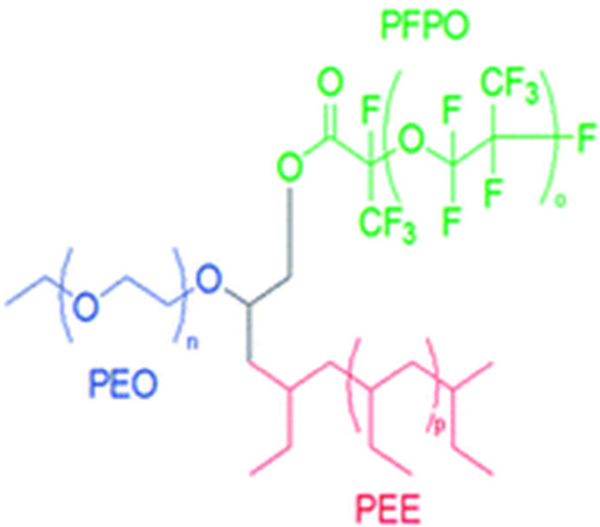


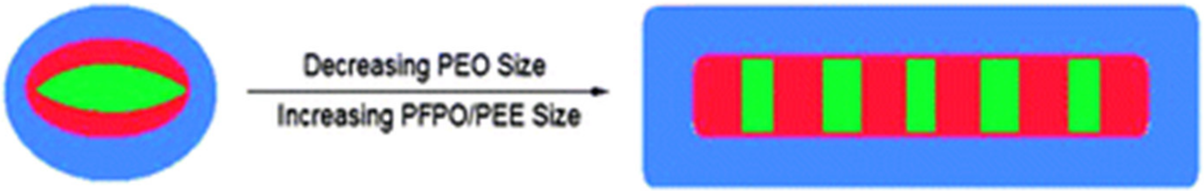
- ABCD miktoarm star polymers (rarely found in the literature due to stringent reaction conditions)
- made by simply coupling two diblock copolymers together through azide–alkyne “click” chemistry

- **1st diblock**: PS synthesized by living anionic polymerization
- chain end-functionality was modified through a series of steps to yield a bifunctional polymer containing an alkyne and 2-bromoisobutyl bromide
- ATRP with *tert*-butyl acrylate yielded the diblock copolymer PS-*b*-PtBA with an alkyne at the junction
- **2nd diblock**: living anionic polymerization of Polyimide (PI)
- Ethylene oxide was directly polymerized from this macroinitiator (PI-Eto)
- Finally, these two diblock copolymers were coupled with an azide–alkyne “click” reaction

Polym. Chem., 2008, **46** , 2154

Self-assembly





- ➢ one hydrophilic PEO arm and two immiscible hydrophobic PEE and PFPO arms

- ➢ **Changes in morphology from varying polymer arm length**

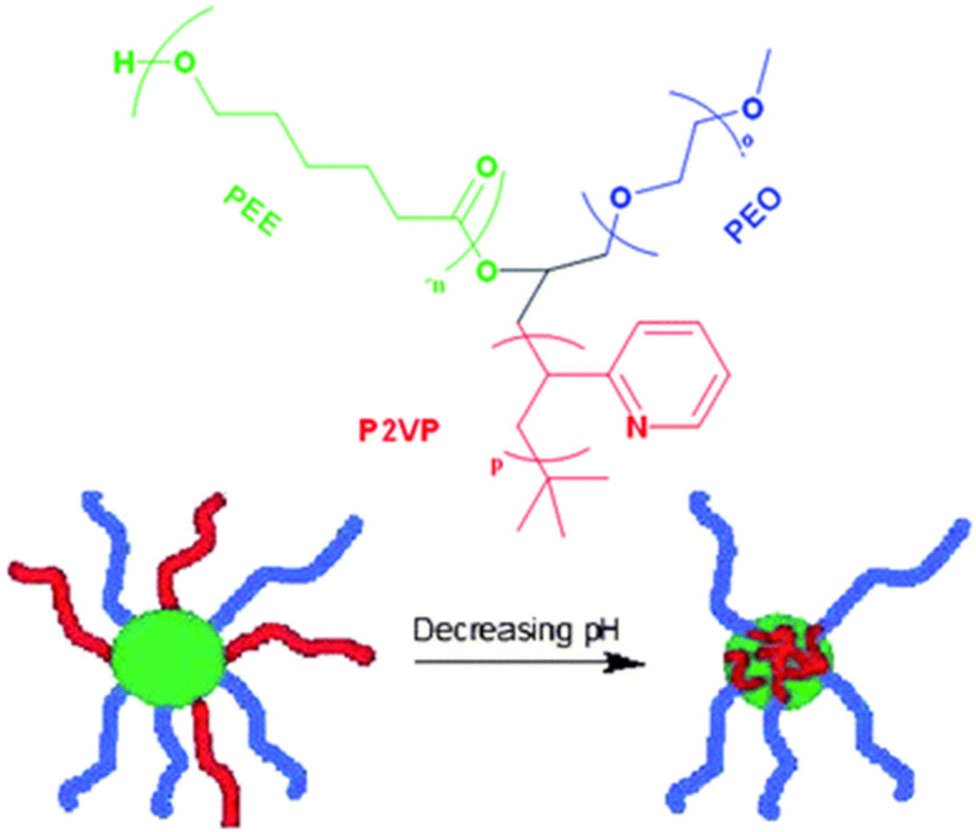
- ➢ ABC miktoarm polymers were made by a “core-first” technique using living anionic polymerization methods
  - poly(ethylene oxide) (PEO),
  - polyethylethylene (PEE), and
  - poly(perfluoropropylene oxide) (PFPO)

- ➢ reduction of the size of PEO arms led to transitions from so-called “hamburger” micelles to a mixture of hamburger micelles and worm-like structures in aqueous solution.

- ➢ short hydrophilic arms of many micelles were believed to group together and create a shared PEO corona so as to best protect the larger hydrophobic inner domains

Science, 2004, **306** , 98

pH-induced changes in morphology

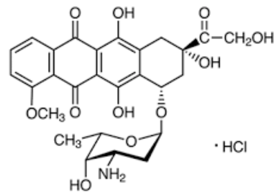


- P2VP arm (red) shifts from the core to the corona with decreasing pH in aqueous media

- ABC miktoarm star composed of:
- PEO, PCL, and pH-responsive P2VP (poly(2-vinyl pyridine))
- in acidic solution: micelle with a core composed of PCL and a mixed corona composed of PEO and protonated P2VP
- Addition of NaOH: deprotonation of the P2VP block
- neutral P2VP could no longer favorably interact with the surrounding solution, the P2VP block in basic conditions collapsed from the corona into the core
- micelle with a corona composed of only PEG and a core composed of PCL and P2VP

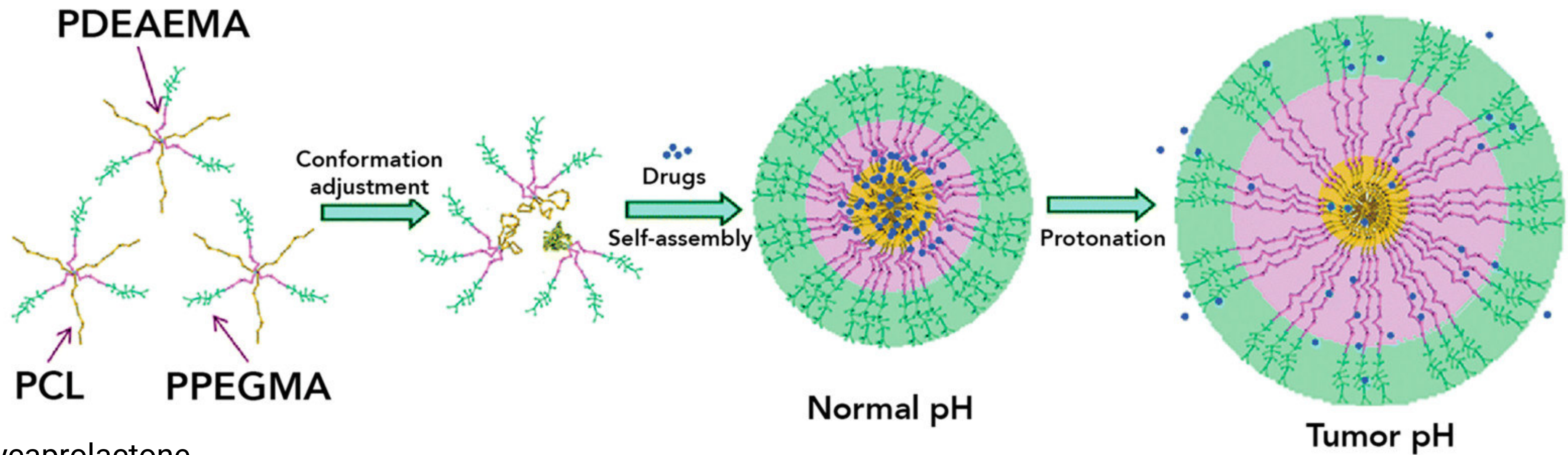
Langmuir, 2009, **25** , 107

pH-induced changes in morphology: Anwendung



Doxorubicin = Chemotherapy

poly[2-(diethylamino)ethyl methacrylate]



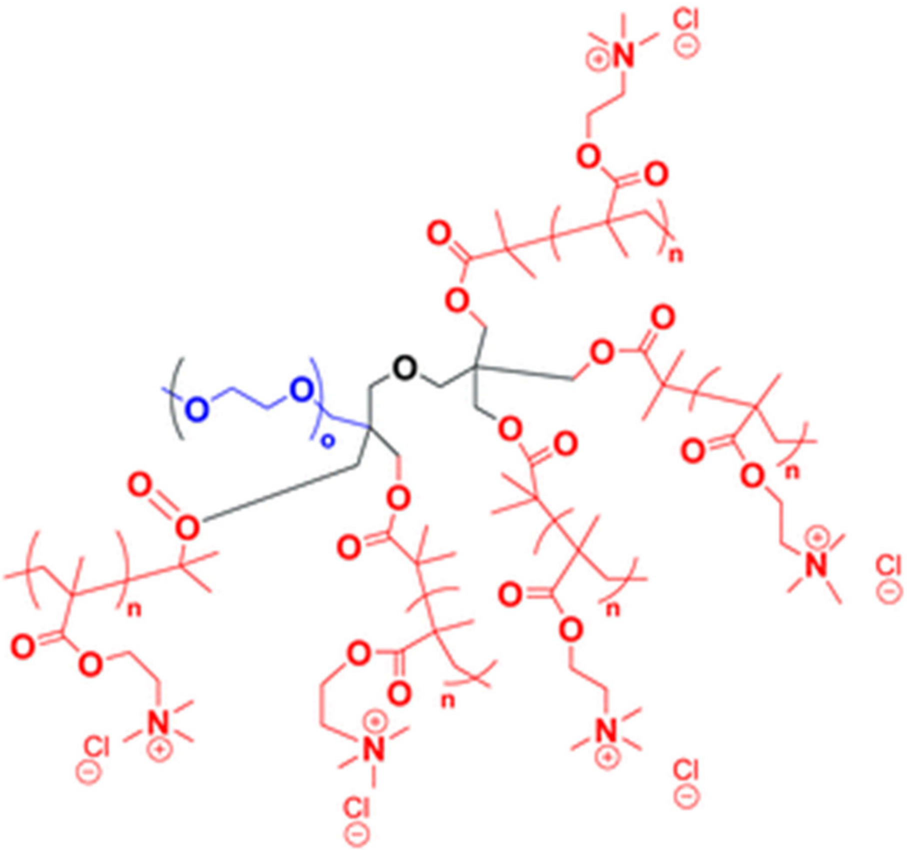
polycaprolactone

poly(poly(ethylene glycol) methacrylate)

- Encapsulation of doxorubicin into $A_3(BC)_3$ µ-star polymers and pH-dependent drug release in tumor cells.

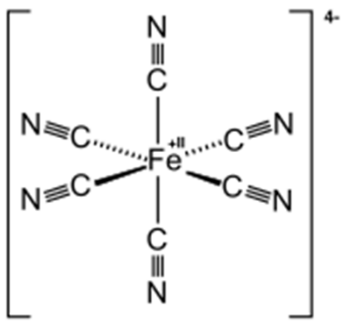
J. Mater. Chem. B 2014, *2* (25), 4008– 4020,

Electrolysis-induced micellization



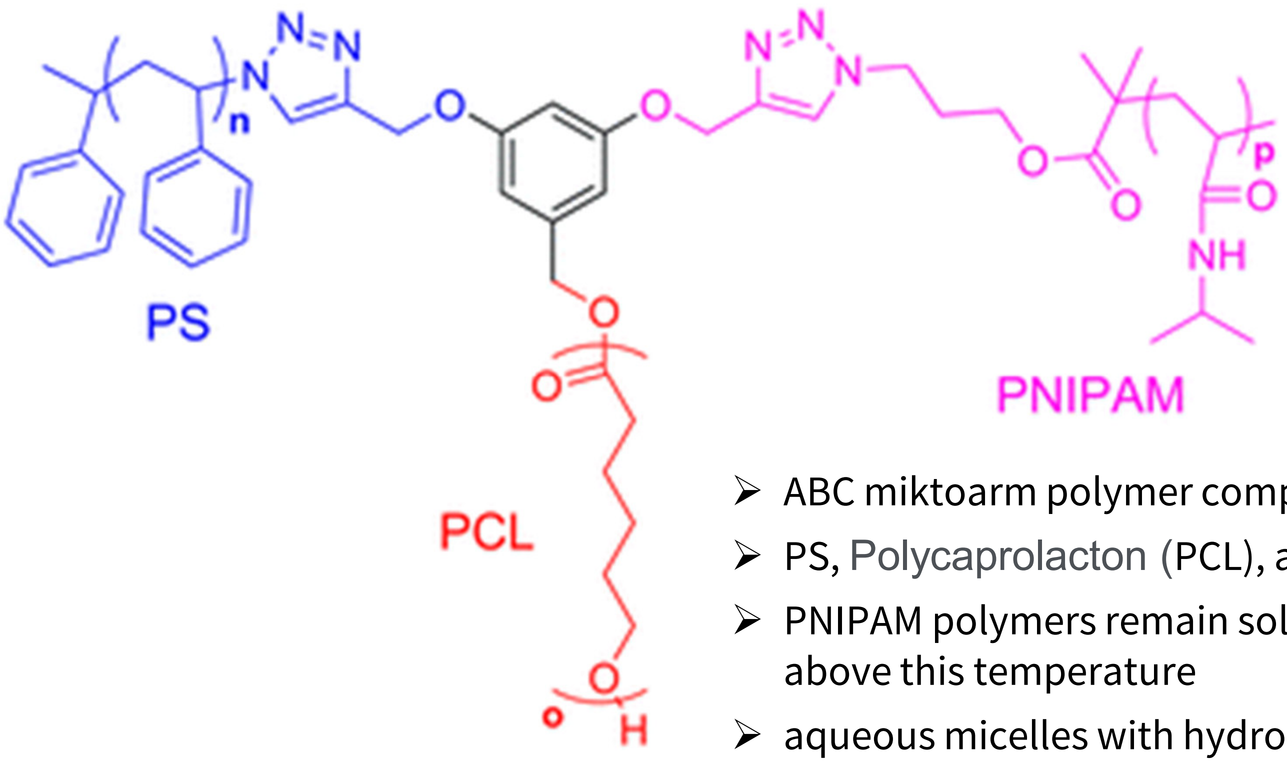
Macromolecules, 2009, **42** , 7254

- micelles composed of miktoarm polymers with one longer PEG arm and 5 shorter arms of poly[{2-(methacryloyloxy)ethyl}trimethylammonium chloride] (PMOTAC)
- PMOTAC is a polyelectrolytepolymer that reacts in the presence of certain counterions
- At high enough concentration of the counterion in aqueous micellar solution, PMOTAC was expected to become hydrophobic, thus driving a phase transition from unimers to micelles.
- Indeed, micellization of this miktoarm polymer was induced in an aqueous solution of this polymer with a specific molar ratio of hexacyanoferrate(III)/hexacyanoferrate(II) counterions at a specific concentration.



Polym. Sci., Part A: Polym. Chem., 2009, **47** , 1636

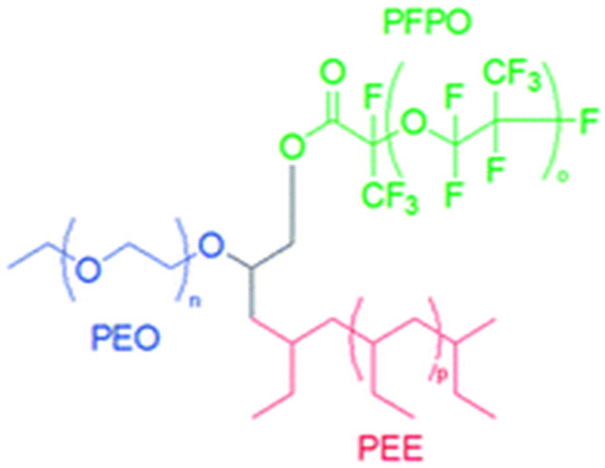
Thermoresponsive micelles

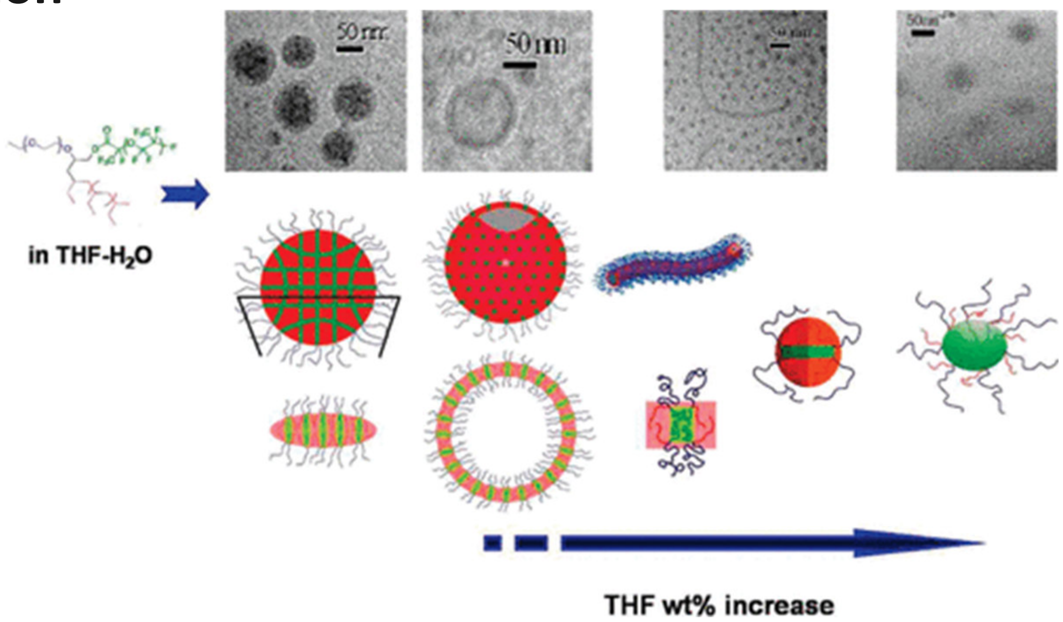


- ABC miktoarm polymer composed of:
- PS, Polycaprolacton (PCL), and poly(*N*-isopropylacrylamide) (PNIPAM)
- PNIPAM polymers remain soluble in water below 32 °C (LCST) and insoluble above this temperature
- aqueous micelles with hydrophilic PNIPAM coronas and mixed hydrophobic PCL and PS cores at room temperature
- At 45 °C, the PNIPAM coronas collapse upon themselves due to their decreasing solubility with the surrounding solvent.
- ~10 nm size decrease of these micelles was observed by DLS

Solvent-induced changes in micellization

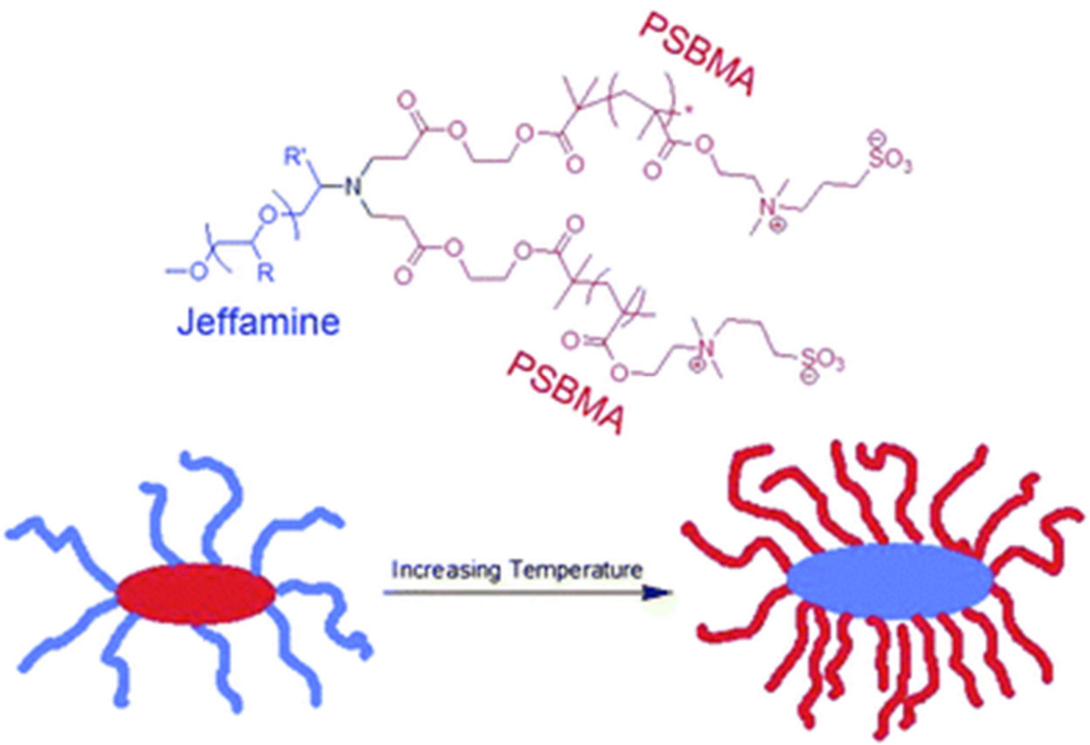
Langmuir 2008, 24, 20, 12001−12009





- *Previously*: decreased PEO arm length: transition from micelles to segmented worms to vesicles
- gradual change in composition of a solvent mixture THF/H_2O could induce this transition
- Tetrahydrofuran (THF) is a solvent selective for PEE and PEO blocks /water is only selective for PEO
- in pure water: morphologies resembling multicompartment disks
- In increasing THF/H_2O mixtures, a transition to: smaller disks / then a mixture of worms and spherical micelles /finally an oblate ellipsoid-shaped micelle in pure THF
- (60 wt% THF) the PEE arm seemed to shift completely from the core to the corona based as this arm became fully solvated by the surrounding solvent mixture, thus forming oblate ellipsoid-shaped micelles

Schizophrenic micelles

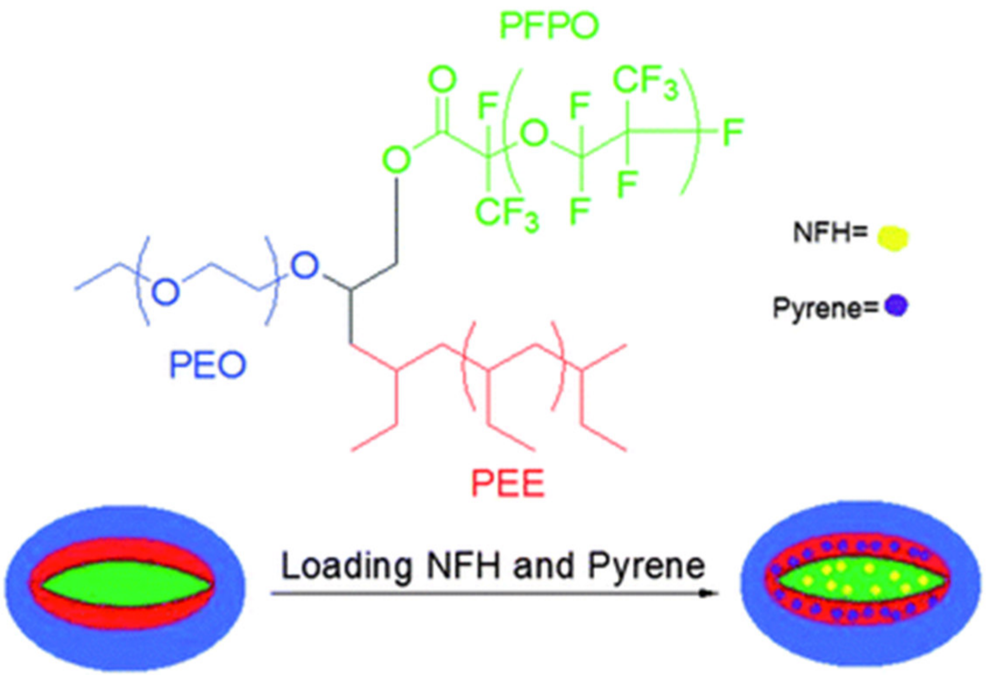


- characterized by a complete inversion of the polymers composing the core and corona in response to external stimuli including temperature, pH, ionic strength, *etc.*

- AB_2 miktoarm polymer composed of two rarely studied arms:
  - one Jeffamine arm (a statistical copolymer of ethylene oxide and propylene oxide)
  - two poly(sulfobetaine methacrylate) (PSBMA) arms
- PSBMA homopolymer is known to be hydrophobic at temperatures below the UCST
- Consequently, at low temperatures of aqueous solution of this AB_2 miktoarm star, micelles were formed with PSBMA core and Jeffamine coronas
- above the UCST of PSBMA, complete dissolution of these polymer arms resulted in unimers in solution as both arms are dissolved.
- As the temperature was further increased, the critical micelle temperature was reached at which point micelles were again formed, but this time with Jeffamine cores and PSBMA coronas

Macromolecules, 2004, **37** , 9728

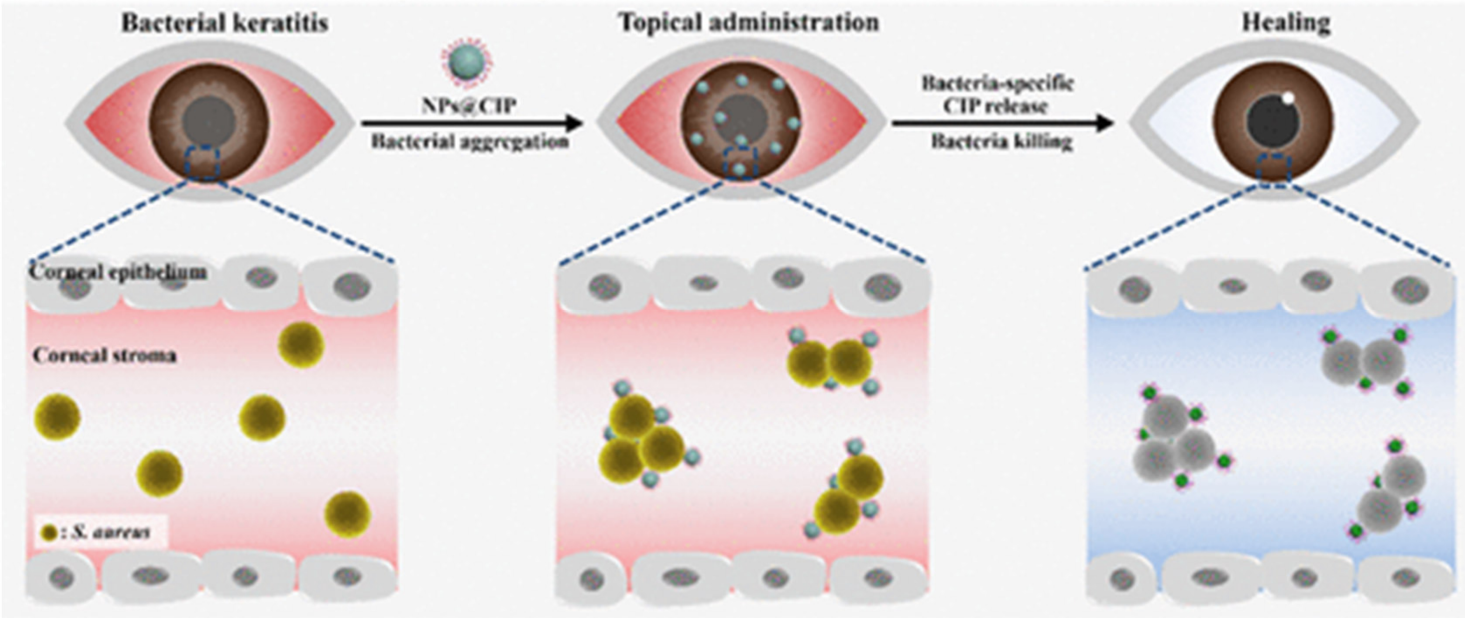
Small molecule storage and release



- ABC-type star composed of PEE, PFPO, and PEO
- simultaneous and segregated storage of two immiscible types of dyes within the multicompartment core of aqueous micelles
- the two hydrophobic dye molecules, pyrene and 1-naphthyl perfluoroheptanyl ketone (NFH) selectively associate with PEE and PFPO respectively in a block copolymer mixture of PEE/PEO and PFPO/PEO,
- it was believed that an aqueous micelle would similarly sequester these two dyes into separate compartments

J. Am. Chem. Soc., 2005, **127** , 17608

Targeted drug release (Bakerielle Keratitis)

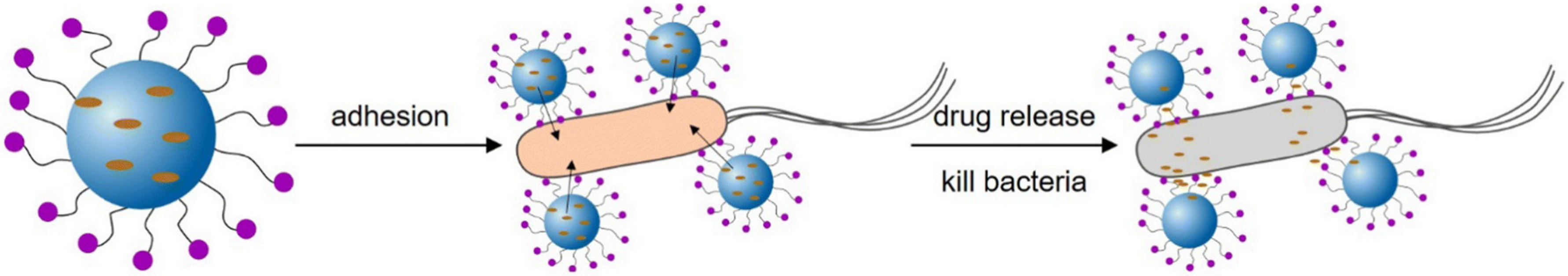


- Bacterial keratitis is an infection of the cornea with bacteria such as *P. aeruginosa* or *S. aureus*
- If left untreated, can cause blindness
- Poor epithelial penetration and a short corneal retention time is a common challenge when administering drugs to treat eye infection

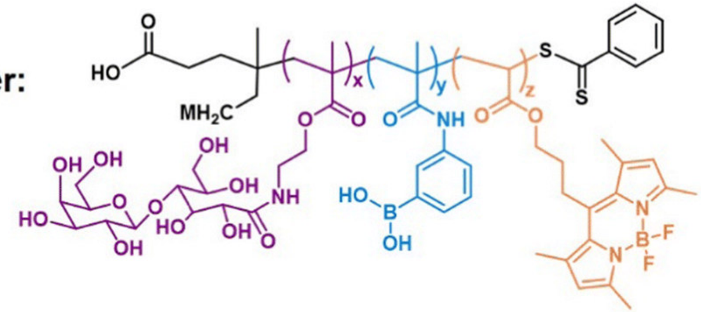
Biomacromolecules, 2021, **22** , 2020 —2032

Biomacromolecules, 2021, **22** , 2020 —2032

Targeted drug release (Bakerielle Keratitis)

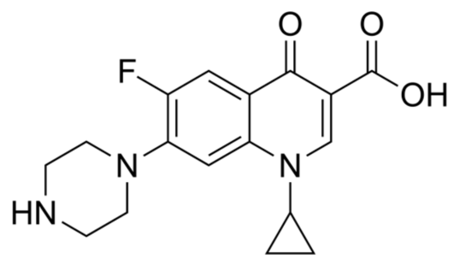


polymer:



- **Epithelium-penetrable polymer micelles**

Ciprofloxacin = synthetisches Breitband Antibiotikum



- Glycopolymers were chosen due to their bioadhesive properties
- Enabling prolonged contact and improved drug accumulation at the target site
- Penetration deep into the cornea
- Efficient cellular internalization of the micelles was demonstrated due to high affinity of boric acid groups for the diol-containing bacterial cell wall
- Enhanced drug penetration and retention inside the pathogenic bacteria